Vapnik–Chervonenkis (VC) Dimension

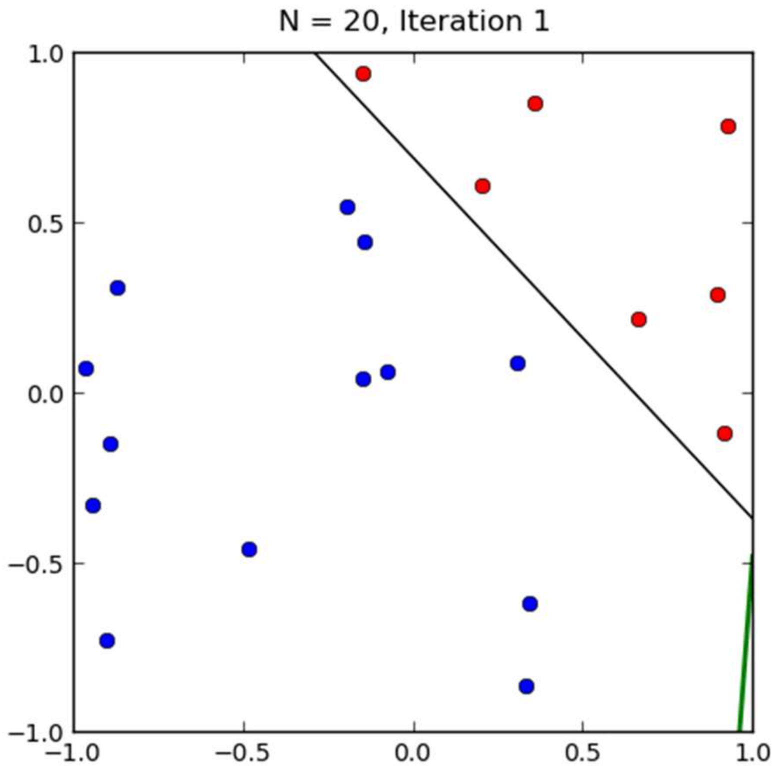
Perceptron Learning

Hypothesis: $h(\boldsymbol{x}) = \text{sign}(\mathbf{w}^{\mathrm{T}}\mathbf{x})$

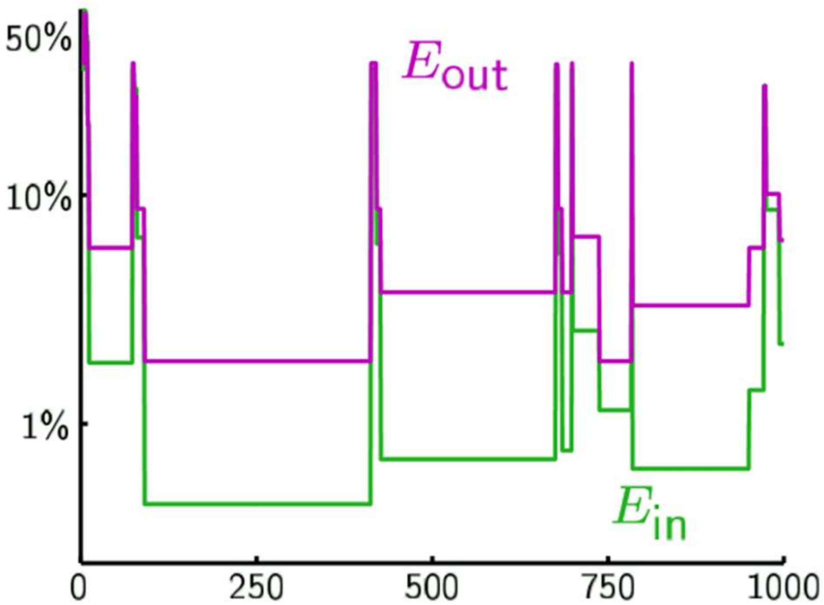
Data: $(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_N, y_N)$

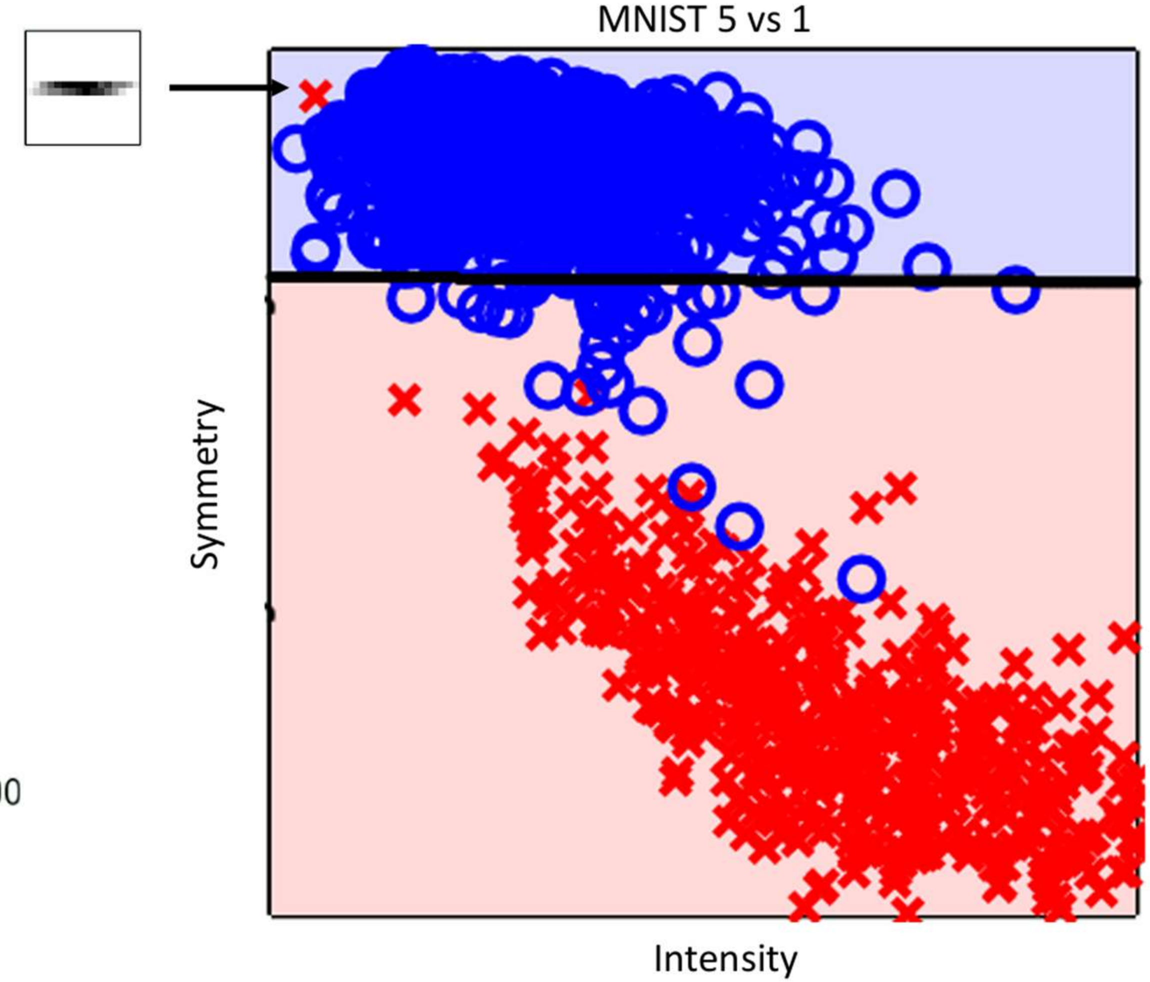
Algorithm:

- Init weights as random
- Pick an $(\boldsymbol{x}_k, y_k)$ that $y_k \neq \text{sign}(\mathbf{w}_{\mathrm{t}}^{\mathrm{T}}\boldsymbol{x}_k)$
- Update $\boldsymbol{w}$: $\boldsymbol{w}_{t+1} = \boldsymbol{w}_t + y_k\boldsymbol{x}_k$

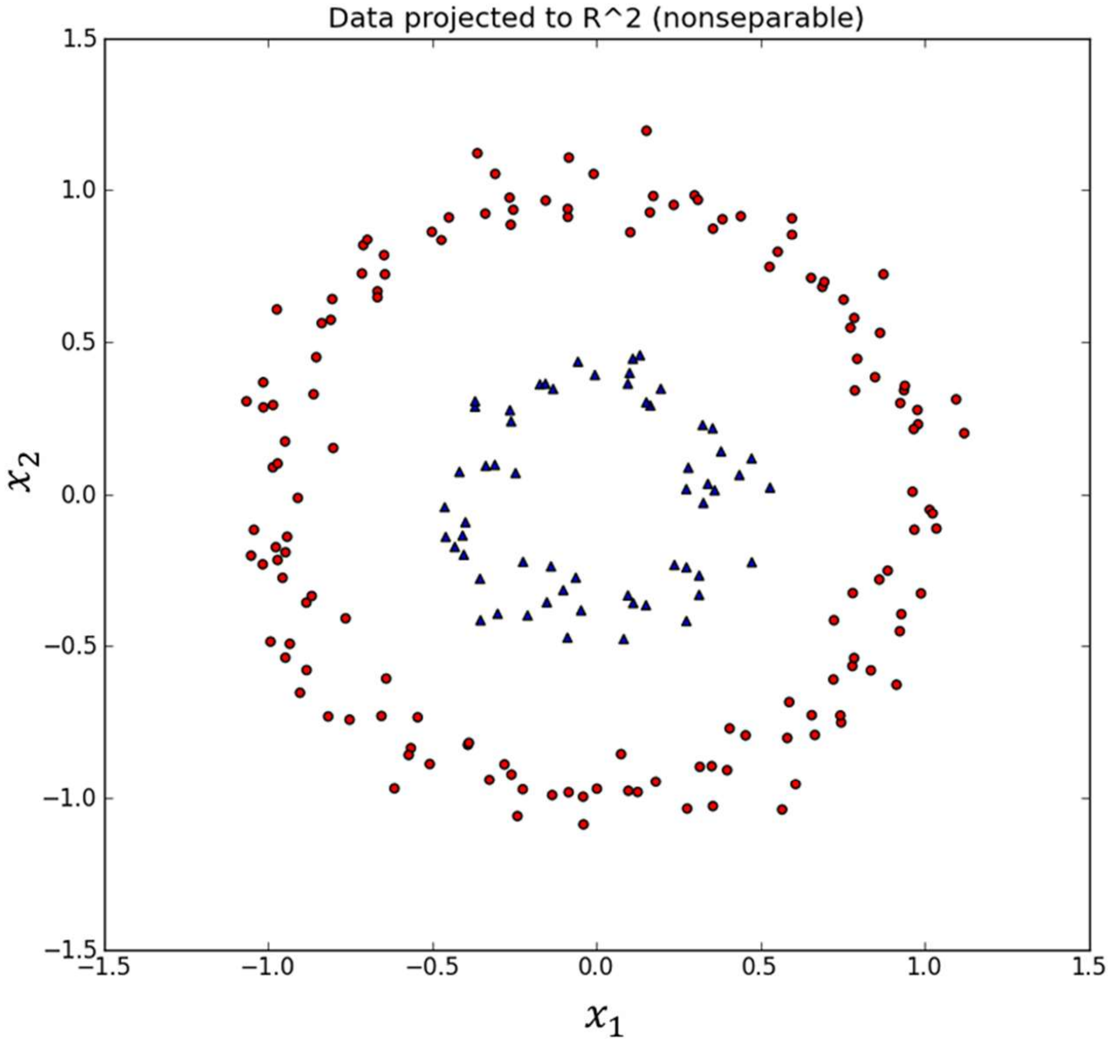


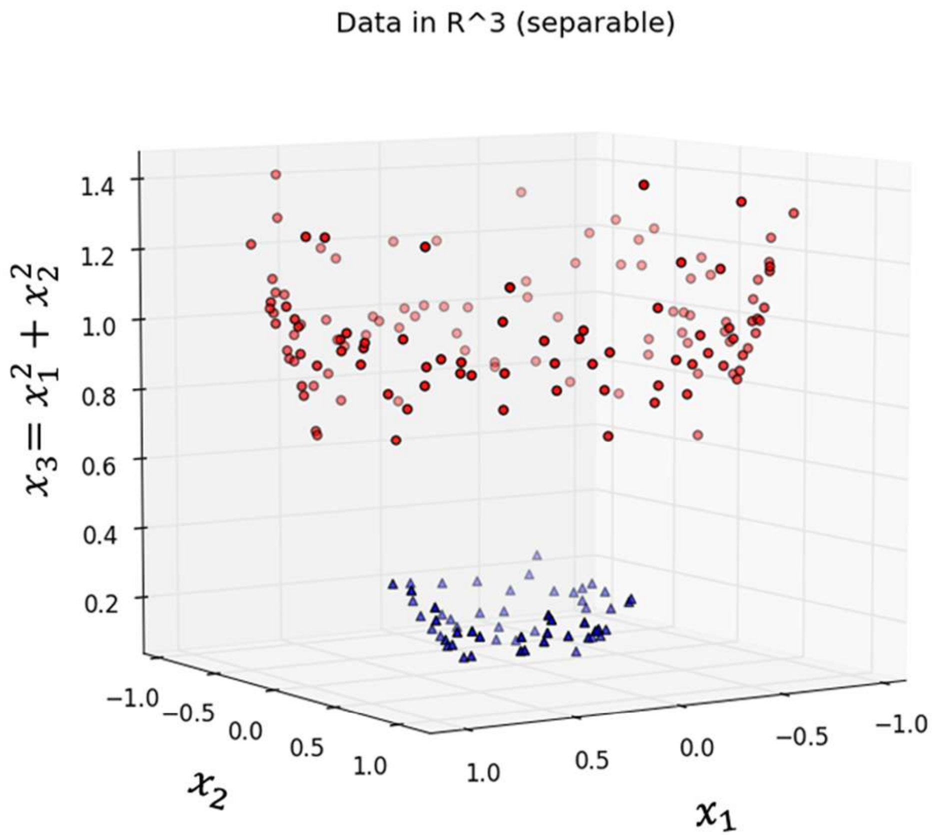
Pocket Algorithm



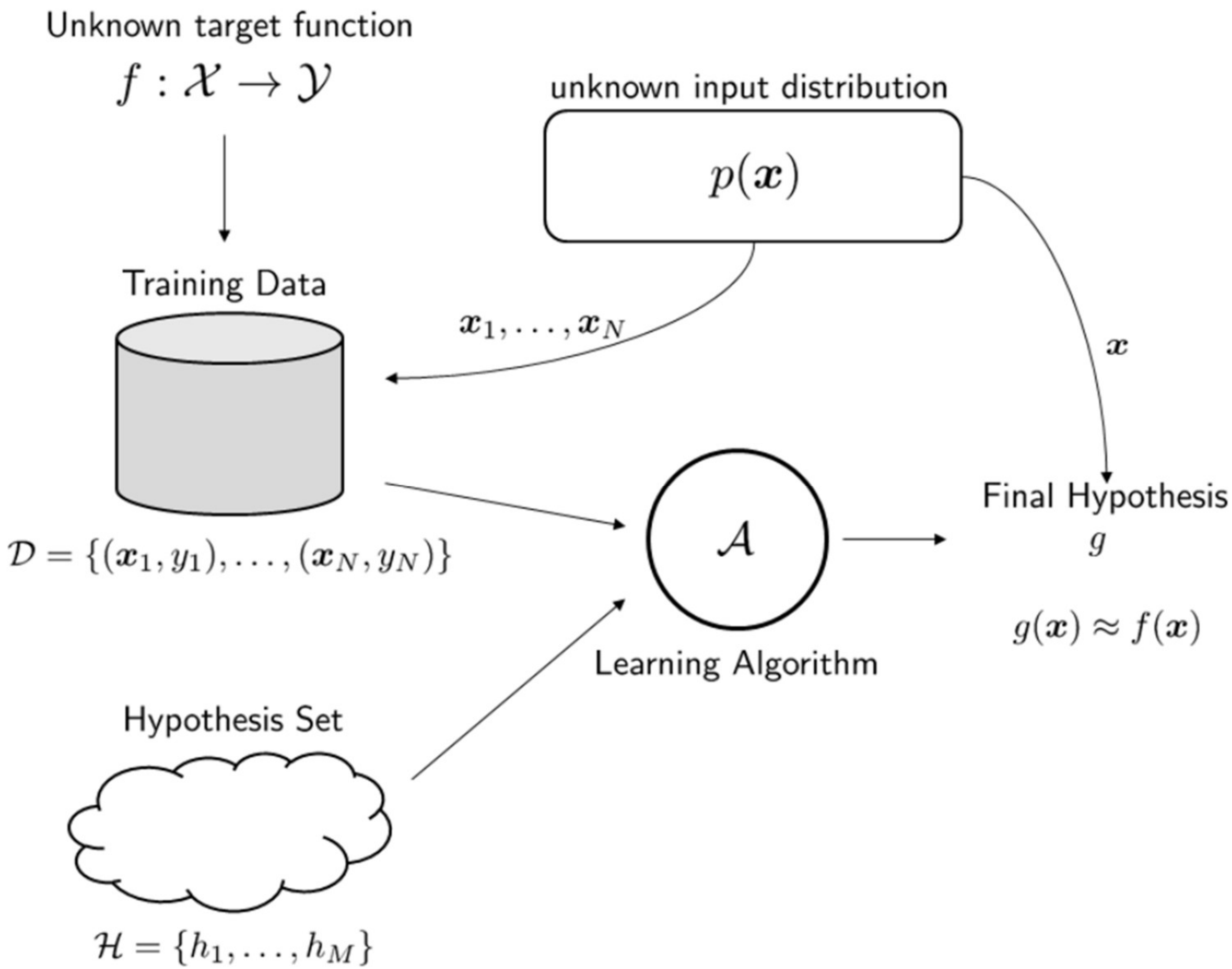


Feature Engineering

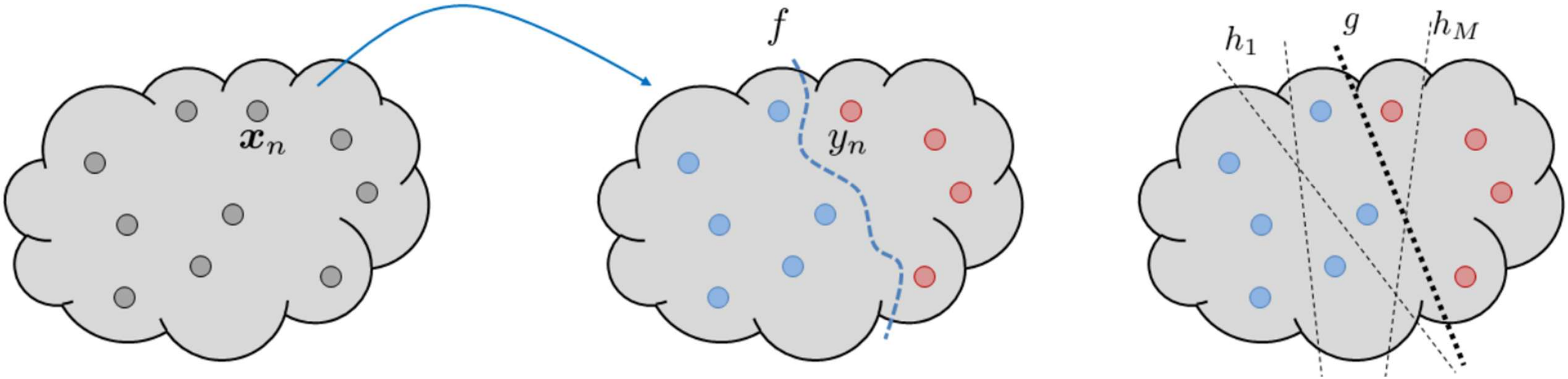




Probabilistic Learning Model



About Hypothesis Set



How can we evaluate our classifier?

Does evaluating on a testing set really tells us how good classifier is?

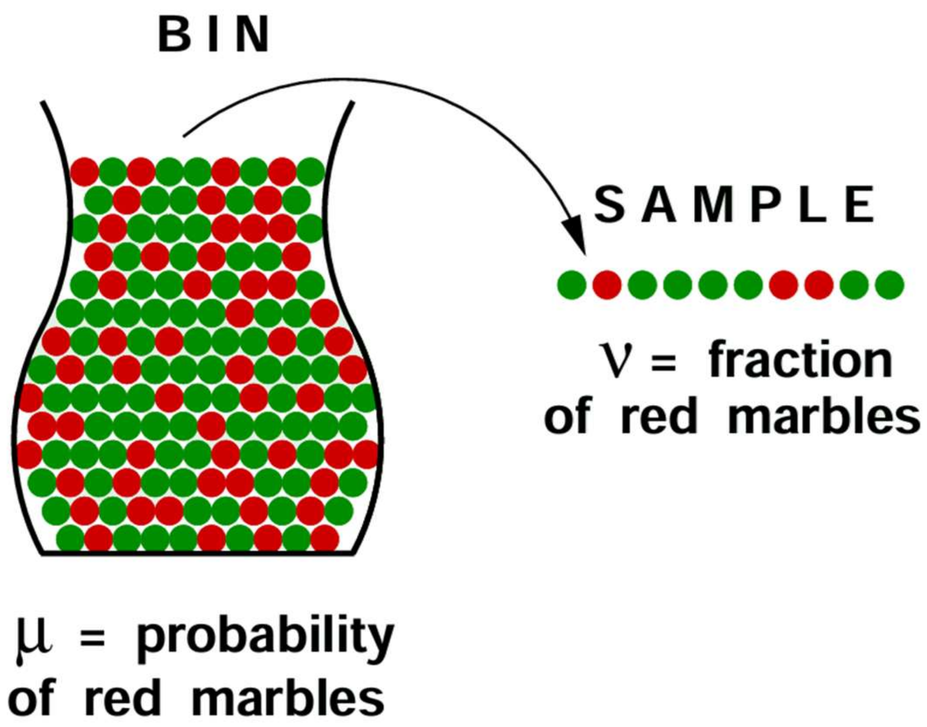
Some idea:

- What does the value of ν tell us about μ?

A random sample from a population tends to agree with the views of the population at large

Let's quantify this.

Hoeffding inequality: $\mathbb{P}[|\nu - \mu| > \epsilon] \leq 2e^{-2\epsilon^2 N}$
for any $\epsilon > 0, N$ − sample size



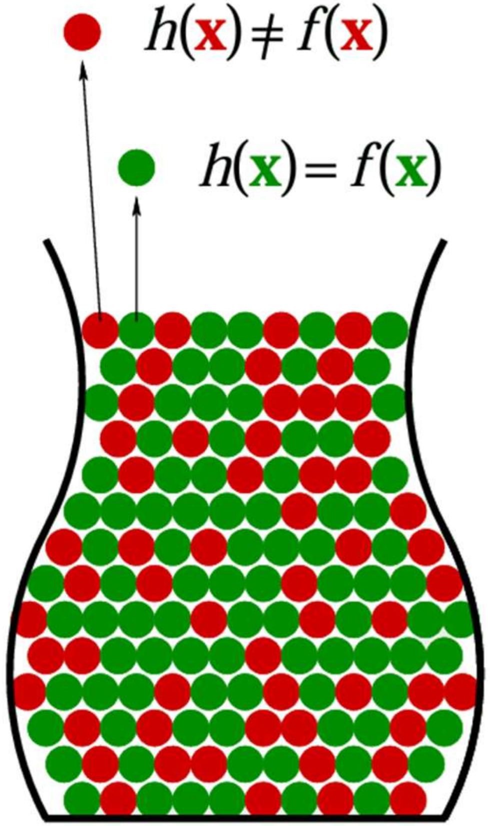
How does it relate to the learning problem?

$E_{in}(h) = \frac{1}{N} \sum_{n=1}^{N} [\![h(\boldsymbol{x}_n) \neq f(\boldsymbol{x}_n)]\!]$ – in-sample error
$E_{out}(h) = \mathbb{E}_{x}[\![h(\boldsymbol{x}) \neq f(\boldsymbol{x})]\!]$ – out-of-sample error

Hoeffding inequality: $\mathbb{P}[|E_{in}(h) - E_{out}(h)| > \epsilon] \leq 2e^{-2\epsilon^2 N}$
for any $\epsilon > 0, N$ – our dataset

Problem: Hoeffding inequality holds for a *fixed* hypothesis function h
i.e. h is already chosen *before* we generate the dataset

But usually we choose g, the final hypothesis, based on our dataset



➢ $[\![h(\boldsymbol{x}_n) \neq f(\boldsymbol{x}_n)]\!]$ is a particular case of $l\big(h(\boldsymbol{x}_n), f(\boldsymbol{x}_n)\big)$

What can we do?

- Suppose that H contains M hypothesis functions $h_1, \dots, h_M$
- The final hypothesis g is one of these potential hypotheses
- To have $|E_{in}(g) - E_{out}(g)| > \epsilon$, we need to ensure that at least one of the M potential hypotheses can satisfy the inequality

$$|E_{in}(g) - E_{out}(g)| > \epsilon \Rightarrow |E_{in}(h_1) - E_{out}(h_1)| > \epsilon$$

$$\textbf{or } |E_{in}(h_2) - E_{out}(h_2)| > \epsilon$$

$$\dots$$

$$\textbf{or } |E_{in}(h_M) - E_{out}(h_M)| > \epsilon$$

This implies that

$$\mathbb{P}[|E_{in}(g) - E_{out}(g)| > \epsilon] \leq 2{\color{red}M}e^{-2\epsilon^2 N}$$

Some Conclusions

Learning goal – to make $E_{out}(g) \approx 0$. But we need two levels of approximation:

$$E_{out}(g) \underbrace{\approx}_{\text{Hoeffding Inequality}} E_{in}(g) \underbrace{\approx}_{\text{Training Error}} 0$$

The first approximation depends from N. The second one requires a good hypothesis and a good learning algorithm.

More complex H:

$$E_{out}(g) \underbrace{\approx}_{\text{Worse if } H \text{ complex}} E_{in}(g) \underbrace{\approx}_{\text{Good if } H \text{ complex}} 0$$

More complex f:

$$E_{out}(g) \underbrace{\approx}_{\text{No effect by } f} E_{in}(g) \underbrace{\approx}_{\text{Worse if } f \text{ complex}} 0$$

Trying to improve the approximation $E_{in(g)} \approx 0$ by increasing the complexity of H needs to pay a price. If H becomes complex, then the approximation $E_{in}(g) \approx E_{out}(g)$ will be hurt.

Generalization Bound

Hoeffding inequality: $\mathbb{P}[|E_{in}(h) - E_{out}(h)| > \epsilon] \leq 2Me^{-2\epsilon^2 N}$

Let $\delta = 2e^{-2\epsilon^2 N}$, then

$$\mathbb{P}[|E_{in}(h) - E_{out}(h)| > \epsilon] \leq \delta \Rightarrow$$

$$\mathbb{P}[|E_{in}(h) - E_{out}(h)| \leq \epsilon] \geq 1 - \delta$$

i.e. with probability $1 - \delta,\ E_{in}(g) - \epsilon \leq E_{out}(g) \leq E_{in}(g) + \epsilon$

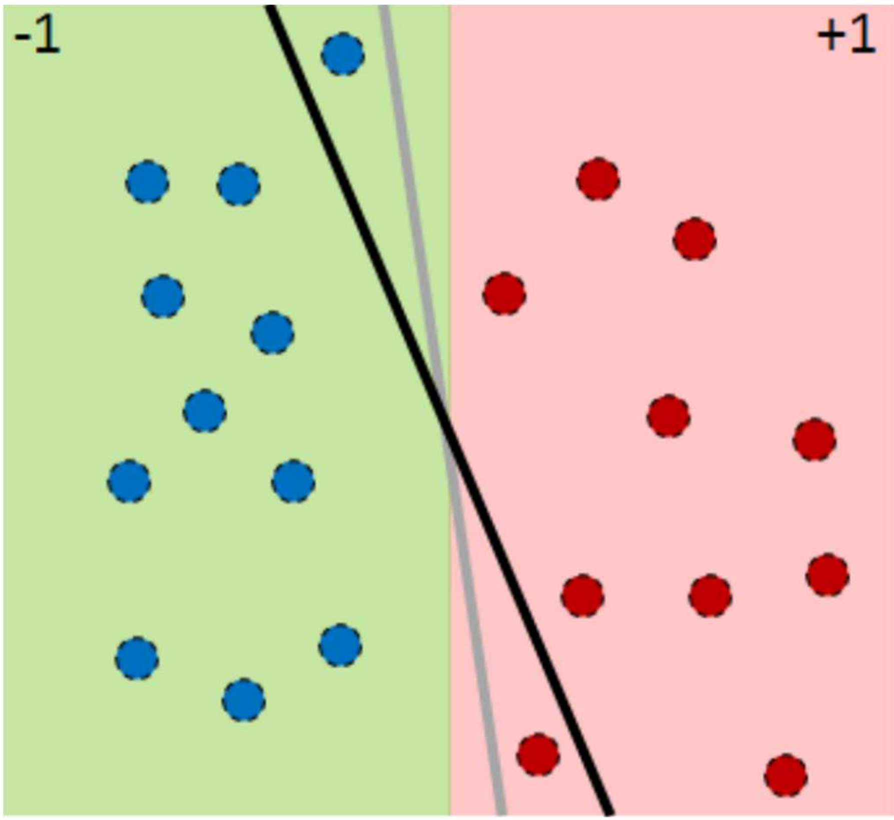
$$\delta = 2Me^{-2\epsilon^2 N} \Leftrightarrow \epsilon = \sqrt{\frac{1}{2N}\log\frac{2M}{\delta}}$$

- The upper bound gives us a safe-guard of how worse $E_{out}(g)$ can be compared to $E_{in}(g)$
- The lower bound tells us what to expect ($E_{out}(g)$ cannot be better than $E\_in(g) - \epsilon$)

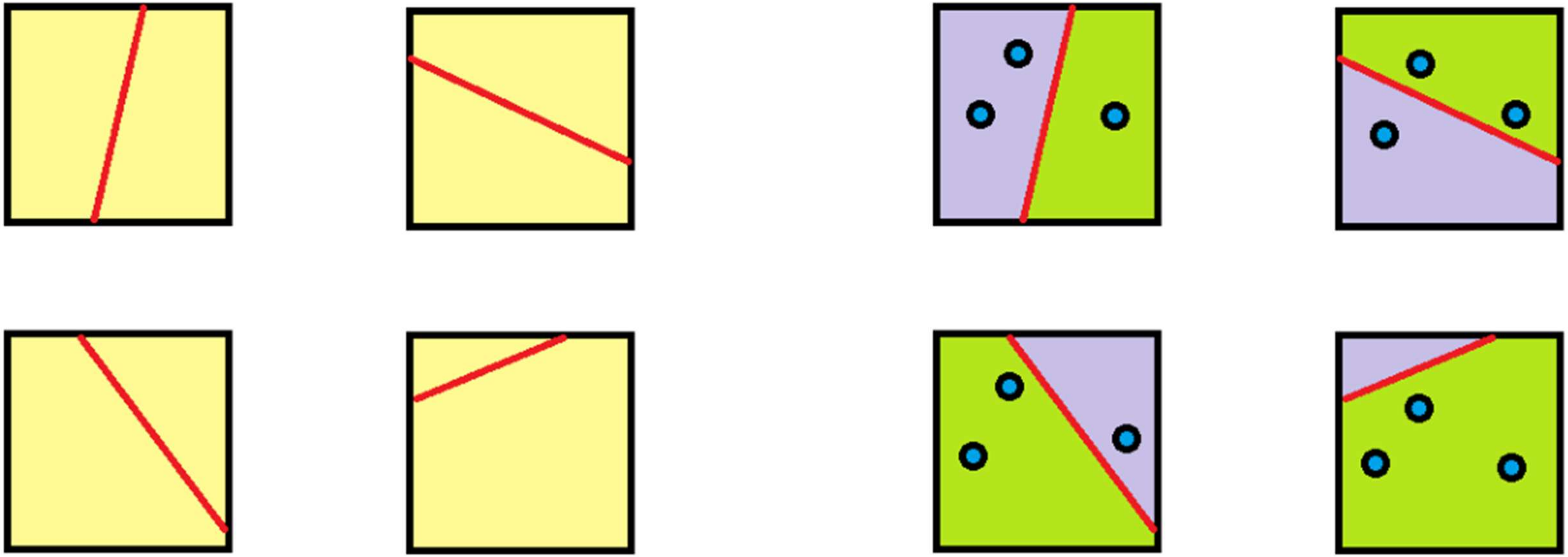
Close Hypotheses

Problem: infinite M

$$\mathbb{P}[|E_{in}(h_1) - E_{out}(h_1)| > \epsilon] \approx \mathbb{P}[|E_{in}(h_2) - E_{out}(h_2)| > \epsilon]$$



Reducing M

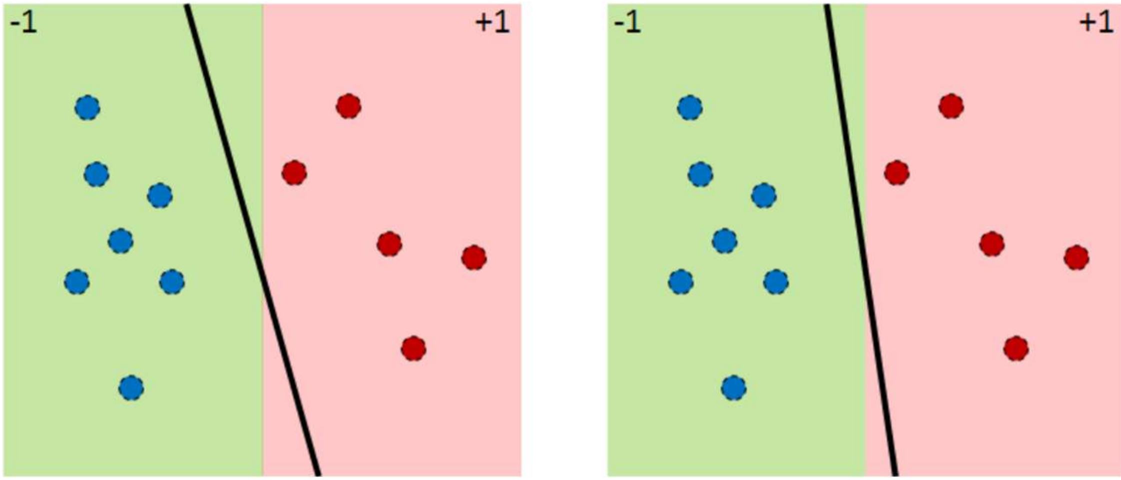


From Hypothesis to Dichotomies

Hypothesis: $h\colon\ X \to \{+1, -1\}$ $-$ for all population samples (number can be infinite)

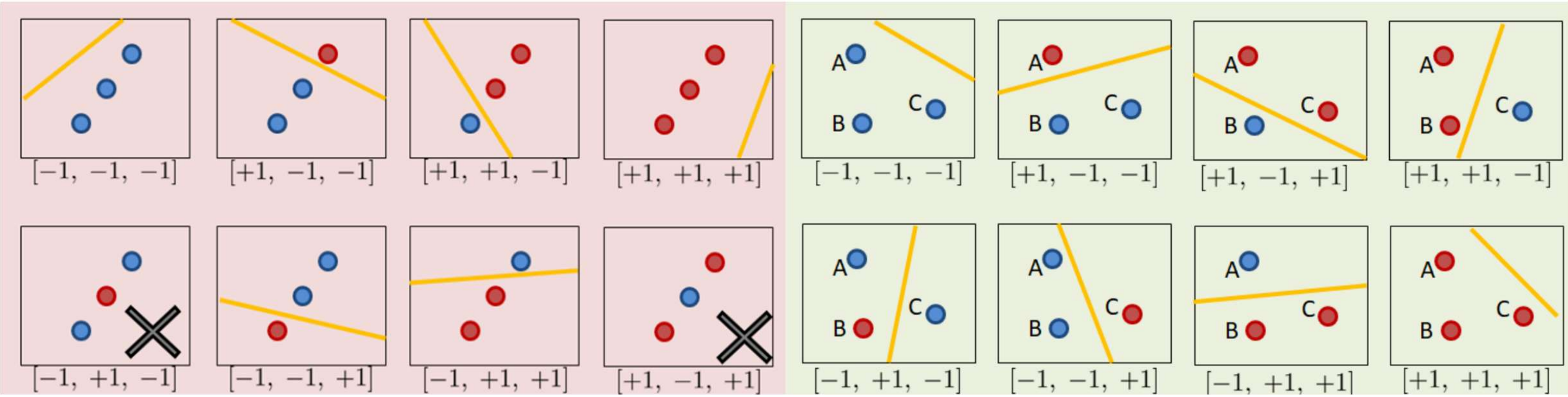
Dichotomy: $h\colon \{\boldsymbol{x}_1, \dots, \boldsymbol{x}_N\} \to \{+1, -1\}$ $-$ for training samples only (number is at most 2^N)

The dichotomies generated by H: $H(x_1, \dots, x_N) = \{(h(x_1), \dots, h(x_N) \mid h \in H\}$



The set of dichotomies is usually much smaller

Counting the Number of Dichotomies



Growth Function

Growth function: $m_H(N) = \max_{x_1,\dots,x_N \in X} |H(\boldsymbol{x}_1, \dots, \boldsymbol{x}_N)|$, $m_H(N) \le 2^N$

Vapnik-Chervonenkis inequality:

$$\mathbb{P}[|E_{in}(g) - E_{out}(g)| > \epsilon] \le 2Me^{-2\epsilon^2 N} \quad \text{(Hoeffding)}$$

$$\Downarrow\Downarrow\Downarrow$$

$$\mathbb{P}[|E_{in}(g) - E_{out}(g)| > \epsilon] \le m_H(2N)4e^{-\epsilon^{\frac{1}{8}}N} \quad \text{(VC)}$$

VC generalization bound:

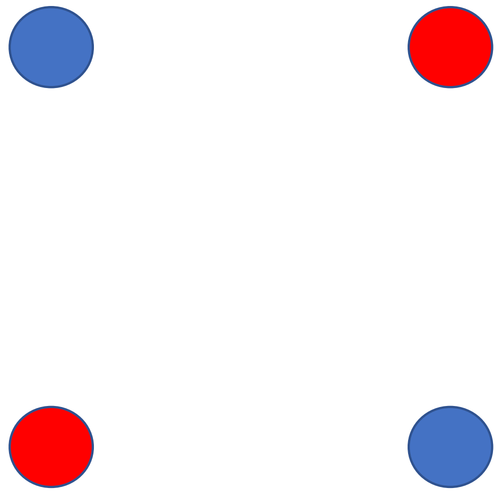
$$E_{in}(g) - \sqrt{\frac{8}{N}\log\frac{4m_H(2N)}{\delta}} \le E_{out}(g) \le E_{in}(g) + \sqrt{\frac{8}{N}\log\frac{4m_H(2N)}{\delta}}$$

with probability $1 - \delta$

Breakpoint

Problem: exponentiality of the growth function $\Longrightarrow \lim_{N\to\infty}\sqrt{\frac{8}{N}\log\frac{4m_{\mathcal{H}}(2N)}{\delta}} \nrightarrow 0$

Breakpoint: $\min(k: m_H(k) < 2^k)$



For linear model in 2D (perceptron):

- Breakpoint: k = 4
- $m_{\mathcal{H}}(4) = 3$

Bounding the Growth Function

What is *shatter*?

➢If a hypothesis set H is able to generate all 2^N dichotomies, then we say that H shatter $x_1, \ldots, x_N$.

$B(N,k)$: max. number of dichotomies on N points so that no subset of size k are shattered

$B(4,2) = 5$

$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$
○	○	○	○
○	○	○	●
○	○	●	○
○	●	○	○
●	○	○	○

$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$
○	○	○	○
○	○	○	●
○	○	●	○
○	●	○	○
●	○	○	○
○	●	●	○

Proof of Polynomiality of a Growth Function in the Presence of a Breakpoint

- $B(N,k) = \alpha + 2\beta$
- $\alpha + \beta \leq B(N-1,k)$
- $\beta \leq B(N-1,k-1)$

Then

$$B(N,k) \leq B(N-1,k) + B(N-1,k-1)$$

Let's prove that

$$B(N,k) \leq \sum_{i=0}^{k-1} C_N^i$$

		# of rows	$\mathbf{x}_1$	$\mathbf{x}_2$	$\dots$	$\mathbf{x}_{N-1}$	$\mathbf{x}_N$
			$+1$	$+1$	$\dots$	$+1$	$+1$
			-1	$+1$	$\dots$	$+1$	-1
	S_1	α	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
			$+1$	-1	$\dots$	-1	-1
			-1	$+1$	$\dots$	-1	$+1$
			$+1$	-1	$\dots$	$+1$	$+1$
			-1	-1	$\dots$	$+1$	$+1$
	S_2^+	β	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
			$+1$	-1	$\dots$	$+1$	$+1$
S_2			-1	-1	$\dots$	-1	$+1$
			$+1$	-1	$\dots$	$+1$	-1
			-1	-1	$\dots$	$+1$	-1
	S_2^-	β	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
			$+1$	-1	$\dots$	$+1$	-1
			-1	-1	$\dots$	-1	-1

Analytic Bound for $B(N,k)$

Sauer's lemma:

$$B(N,k) \le \sum_{i=0}^{k-1} C_N^i$$

Proof: $B(N,1) = 1; B(1,k>1) = 2$. The statement is true for $k = 1$ or $N = 1$.

Assume the statement is true for all numbers before N and all k.

$$B(N+1,k) \le B(N,k) + B(N,k-1) \le \sum_{i=0}^{k-1} C_N^i + \sum_{i=0}^{k-2} C_N^i = \sum_{i=0}^{k-1} C_N^i + \sum_{i=1}^{k-1} C_N^{i-1} =$$

$$= 1 + \sum_{i=1}^{k-1} (C_N^i + C_N^{i-1}) = 1 + \sum_{i=1}^{k-1} C_{N+1}^i = \sum_{i=0}^{k-1} C_{N+1}^i$$

$m_H(N)$ is bounded by $B(N,k)$!

Theorem: suppose that H has a break point at k. Then,

$$m_H(N) \leq B(N,k)$$

- Consider any k points
- They cannot be shattered (otherwise k would not be a break point)
- $B(N,k)$ is largest such list (by definition)
- $m_H(N) \leq B(N,k)$

Once Broken, Forever Polynomial

VC dimension: $d_{VC}(H)$ for hypothesis type H, is the maximum number N,

such that $m_H(N) = 2^N$. If $m_H(N) = 2^N$ for all N, then $d_{VC}(H) = \infty$

$$d_{VC}(H) = k - 1, \text{where } k - \text{is the breakpoint}$$

If k is any break point for H, so $m_H(N) < 2^k$

$$\boxed{m_H(N) \le \sum_{i=0}^{k-1} C_{N+1}^{i} = \sum_{i=0}^{d_{VC}} C_{N+1}^{i} \le N^{d_{VC}} + 1}$$

VC dimension of Perceptron

Consider the input space $X = \mathbb{R}^d \cup \{1\}$. The VC dimension of the perceptron algorithm is $d_{VC} = d + 1$.

Part 1: $d_{VC} \geq d + 1$

$$\begin{bmatrix} -\mathbf{x}_1^T- \\ -\mathbf{x}_2^T- \\ \vdots \\ -\mathbf{x}_{d+1}^T- \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ 1 & 0 & 1 & & 0 \\ & & & \ddots & 0 \\ 1 & 0 & \dots & 0 & 1 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{d+1} \end{bmatrix} = \begin{bmatrix} \pm 1 \\ \pm 1 \\ \vdots \\ \pm 1 \end{bmatrix}$$

Part 2: $d_{VC} \leq d + 1$

$$\boldsymbol{x}_{d+2} = \sum_{j=1}^{d+1} \alpha_j \boldsymbol{x}_j, \qquad \boldsymbol{w}^T \boldsymbol{x}_{d+2} = \sum_{j=1}^{d+1} \alpha_j \boldsymbol{w}^T \boldsymbol{x}_j, \qquad y_i = \begin{cases} \text{sign}(\alpha_i), & i \neq d+2 \\ -1, & \text{otherwise} \end{cases}$$

Theory Versus Practice

The VC analysis allows us to reach outside the data for general H

- a single parameter characterizes complexity of H
- d_{vc} depends only on H
- E_{in} can reach outside D to E_{out} when d_{vc} is finite

In Practice ...

The VC bound is loose

- Hoeffding
- $m_H(N)$ is a worst case # of dichotomies, not average case or likely case
- The polynomial bound on $m_H(N)$ is loose

But it's a good guide – models with small d_{vc} are good!

Sample Complexity

Set the error bar at ϵ

$$\epsilon \geq \sqrt{\frac{8}{N} \ln \frac{4\left((2N)^{d_{VC}} + 1\right)}{\delta}}$$

Solve for N:

$$N \geq \sqrt{\frac{8}{\epsilon^2} \ln \frac{4\left((2N)^{d_{VC}} + 1\right)}{\delta}} = O(d_{VC} \ln N)$$

For $d_{VC} = 3, \epsilon = 0.1, \delta = 0.1$. We will converge to $N \approx 30000$.

But it's gross overestimate!

Roughly $10 \times d_{vc}$ examples needed to get good generalization

The Test Set

Another way to estimate $E_{out}(g)$ is using a test set to obtain $E_{test(g)}$

E_{test} is better than E_{in}: you don't pay the price for fitting

- You can use $M = |H| = 1$ in the Hoeffding bound with E_{test}
- The test set should not be used to select g!

Both a test and training set have variance

- The training set has optimistic bias due to selection – fitting the data
- A test set has no bias

The price for a test set is fewer training examples